

HW3_加分

加分題 A

讀檔

```
data_SRBCT <- read.delim("SRBCT_train.txt")
data1 <- as.matrix(data_SRBCT[, -c(1:2)])

# 提取 Image_id
Image_id <- data_SRBCT[,1]
```

用 **normalize** 後的 **SRBCT_train.txt**,分別計算每個 **gene**

在 **EWS / BL / NB / RMS** 四組樣本各自的平均表現量

```
min_max_norm <- function(x){
  (x - min(x)) / (max(x) - min(x))
}

data1_norm <- t(apply(data1, 1, min_max_norm))

# 取得所有欄位名稱
cols <- colnames(data1)
labels <- sub("\\\\..*", "", cols)

# 取出 EWS / BL / NB / RMS 四組樣本各自的資料
EWS_data <- data1_norm[, which(labels == "EWS")]
BL_data <- data1_norm[, which(labels == "BL")]
NB_data <- data1_norm[, which(labels == "NB")]
RMS_data <- data1_norm[, which(labels == "RMS")]

# 四組樣本各自的平均表現量
EWS_data_mean <- apply(EWS_data, 1, mean)
BL_data_mean <- apply(BL_data, 1, mean)
NB_data_mean <- apply(NB_data, 1, mean)
RMS_data_mean <- apply(RMS_data, 1, mean)
```

接著找出符合下列三種表現樣式的 genes

```
# SNR 公式：
# SNR = ( $\mu_{\text{target}}$  -  $\mu_{\text{others}}$ ) / ( $\sigma_{\text{target}}$  +  $\sigma_{\text{others}}$ )
# 其中  $\mu$  為該群樣本在此 gene 的平均值、 $\sigma$  為標準差。
SNR <- function(t_mean, t_sd, o_mean, o_sd){
  (t_mean - o_mean) / (t_sd + o_sd)
}
```

1. BL-low pattern: BL 組的平均低於 EWS、NB、RMS 三組各自的平均

```
bl_low_rows <- which(BL_data_mean < EWS_data_mean &
  BL_data_mean < NB_data_mean &
  BL_data_mean < RMS_data_mean)

bl_low_rows # 觀察
```

Output:

```
> bl_low_rows
[1] 1 2 6 7 11 14 16 17 21 23 30 31 32 33 38 45 47 50 53 63 64 66 68 70 72 73 77 78 79 84 86 88
[33] 89 90 92 93 95 98 99 100 104 107 108 110 111 115 116 117 120 121 122 125 126 127 132 133 135 136 139 140 141 142 146 147
[65] 150 152 153 154 156 160 162 167 169 171 177 182 185 186 187 192 193 197 198 200 201 203 204 216 217 220 221 224 227 230 231 233
[97] 235 236 237 239 244 246 247 248 250 251 253 254 255 256 257 258 259 260 266 269 271 274 275 277 280 281 282 284 290 295 296 297
[129] 303 310 314 319 322 324 326 327 329 331 333 334 336 338 341 349 350 351 352 353 358 359 360 362 363 366 367 371 374 376 380 381
[161] 382 383 385 390 393 395 397 399 401 402 403 412 415 416 417 418 419 423 424 425 426 427 428 430 431 433 434 435 436 442 445 449
[193] 451 452 453 454 455 456 472 476 477 479 482 484 489 492 494 495 496 497 500 504 505 509 510 511 512 521 522 523 524 526 529 531
[225] 532 537 544 545 546 548 551 553 558 560 561 565 566 567 575 576 579 580 583 586 587 590 592 594 607 608 612 616 618 621 624 625
[257] 626 627 633 634 638 648 650 651 653 654 656 657 658 659 660 661 665 666 671 672 673 680 682 687 688 690 693 694 697 699 704 705
[289] 707 710 711 714 715 717 718 719 722 723 726 731 732 737 741 742 749 753 756 757 761 765 766 772 785 786 787 788 789 792 796 799
[321] 800 805 806 807 811 815 818 820 821 827 829 831 834 835 837 841 843 847 850 851 852 853 856 857 858 859 861 865 866 867 876 877
[353] 879 891 894 900 901 902 905 907 909 910 913 917 919 920 923 929 930 931 933 936 941 945 946 947 949 951 952 958 959 960 961 966
[385] 968 969 972 974 980 981 983 984 986 987 989 991 995 999 1005 1006 1009 1010 1011 1013 1022 1029 1030 1033 1046 1049 1050 1055 1056 1060 1066 1069
[417] 1071 1073 1076 1078 1082 1083 1084 1085 1086 1088 1089 1100 1103 1104 1105 1107 1110 1111 1112 1113 1115 1119 1120 1121 1123 1124 1125 1128 1138 1148 1151 1154
[449] 1156 1159 1160 1164 1170 1173 1175 1184 1188 1189 1191 1192 1194 1195 1197 1201 1202 1203 1205 1207 1208 1210 1217 1219 1220 1223 1225 1226 1227 1233 1235 1236
[481] 1237 1238 1239 1240 1243 1244 1245 1246 1250 1259 1260 1269 1270 1277 1281 1285 1292 1296 1297 1301 1304 1305 1311 1313 1316 1319 1327 1329 1332 1334 1336 1337
[513] 1339 1340 1342 1344 1345 1346 1347 1349 1350 1352 1353 1354 1355 1360 1361 1363 1365 1367 1373 1374 1377 1389 1392 1393 1396 1398 1408 1415 1419 1423 1427 1430
[545] 1431 1434 1435 1437 1438 1440 1442 1446 1464 1465 1468 1475 1479 1480 1481 1482 1484 1485 1570 1577 1578 1581 1583 1584 1588 1590 1593 1594 1595 1599 1601 1604 1608
[577] 1533 1538 1539 1544 1545 1547 1548 1549 1550 1551 1554 1555 1556 1557 1561 1564 1565 1570 1577 1578 1581 1583 1584 1588 1590 1593 1594 1595 1599 1601 1604 1608
[609] 1610 1612 1613 1619 1620 1625 1626 1627 1628 1635 1636 1642 1643 1644 1645 1646 1647 1655 1656 1657 1662 1663 1664 1668 1673 1675 1677 1693 1696 1697 1700 1702
[641] 1706 1707 1708 1711 1714 1717 1721 1722 1723 1724 1727 1732 1734 1738 1739 1745 1746 1748 1749 1750 1751 1753 1754 1755 1756 1757 1760 1761 1763 1764 1766 1768
[673] 1771 1772 1773 1777 1782 1788 1789 1792 1795 1804 1806 1808 1809 1810 1811 1813 1814 1815 1816 1821 1822 1824 1825 1828 1830 1831 1834 1837 1841 1842 1846 1847
[705] 1853 1855 1856 1858 1859 1868 1869 1872 1875 1876 1879 1880 1881 1886 1888 1889 1890 1895 1896 1897 1908 1910 1912 1920 1929 1937 1938 1942 1943 1944 1959 1961
[737] 1966 1969 1971 1975 1978 1980 1982 1984 1986 1987 1990 1991 1994 1997 1998 2000 2002 2004 2006 2007 2011 2012 2018 2019 2020 2022 2023 2024 2029 2031 2032 2033
[769] 2040 2041 2046 2049 2051 2052 2054 2057 2059 2061 2070 2071 2073 2077 2078 2079 2081 2082 2087 2088 2089 2090 2092 2093 2094 2095 2096 2099 2101 2102 2105 2108
[801] 2110 2111 2114 2120 2123 2126 2130 2132 2134 2137 2138 2141 2144 2146 2148 2149 2157 2159 2162 2163 2166 2168 2170 2178 2179 2183 2185 2187 2190 2195 2196 2198
[833] 2199 2200 2201 2202 2205 2207 2209 2211 2214 2215 2217 2219 2220 2221 2225 2226 2227 2231 2233 2235 2236 2238 2245 2247 2250 2251 2252 2253 2254 2257 2259 2260
[865] 2263 2265 2272 2273 2281 2282 2283 2284 2285 2286 2288 2289 2290 2291 2293 2294 2297 2299 2302 2303 2304 2306
```

過濾:只留下 BL 組的平均低於 EWS、NB、RMS 三組各自平均的資料

```
bl_low_data <- data1_norm[bl_low_rows, ]
```

對每個 gene 計算 SNR (signal-to-noise ratio), 並挑出 SNR 值最小的那個 gene。

```
# target 為過濾後 BL 類別的資料，其餘為 others
# 計算 target、others 的 mean、sd，以計算 snr
target1 <- bl_low_data[, which(labels == "BL")]
target1 # 觀察
target1_mean <- apply(target1, 1, mean)
target1_sd <- apply(target1, 1, sd)

others1 <- bl_low_data[, -which(labels == "BL")]
others1 # 觀察
others1_mean <- apply(others1, 1, mean)
others1_sd <- apply(others1, 1, sd)

snr1 <- SNR(target1_mean, target1_sd, others1_mean,
             others1_sd)
idx1 <- which.min(snr1) # 索引相對於 bl_low_data
```

Output:

	BLC5	BLC6	BLC7	BLC8	BLC1	BLC2	BLC3	BLC4
1	0.064990700	0.033059016	0.0102023561	0.070317344	0.0212502114	0.0791950848	0.0877346260	0.000000000
2	0.018397056	0.042073268	0.0266624007	0.000000000	0.0036794113	0.0033594625	0.0077854210	0.037807284
3	0.037731004	0.018205807	0.0569285670	0.063428111	0.1782238712	0.1485951599	0.1582913460	0.123464771
4	0.055292609	0.171769715	0.1599989583	0.101861316	0.0755528935	0.1531370238	0.0767898646	0.106854773
5	0.157929279	0.169503250	0.2848527312	0.272152404	0.1821237572	0.2304417621	0.1274644577	0.176917687
6	0.431925115	0.332846318	0.2707642019	0.087340894	0.1507107127	0.2513496112	0.4335347432	0.217374078
7	0.106994419	0.149982187	0.1104381902	0.145152991	0.0721212841	0.1659343704	0.2362348098	0.211930491
8	0.240877712	0.289940828	0.1693786982	0.179240631	0.1306706114	0.2409270217	0.2225838264	0.163412229
9	0.097787976	0.098741879	0.1477518820	0.000000000	0.2439929875	0.4348767660	0.2045477983	0.034211612
10	0.270860773	0.309444487	0.1837762454	0.047648563	0.4209343767	0.3873092121	0.2813202138	0.222514914
11	0.205080630	0.021801413	0.0391825152	0.125733308	0.1422412617	0.0680509700	0.0879150864	0.000000000
12	0.230202713	0.343298819	0.0426571387	0.000000000	0.1848476010	0.1072626513	0.1766078460	0.180545428
13	0.184151551	0.065935990	0.0758456475	0.133132572	0.1424119336	0.1471741719	0.2670705231	0.000000000
14	0.016313636	0.012714278	0.0148894298	0.015407323	0.0234346678	0.0171163706	0.0265161324	0.020534466
15	0.268093984	0.083105658	0.1931349660	0.075604789	0.0692464358	0.1450499230	0.3081317371	0.333565148
16	0.098053148	0.123419071	0.1366349297	0.088993889	0.0893491545	0.1831391218	0.1091374165	0.178911468
17	0.492001920	0.430876590	0.5426497641	0.450091978	0.3841977925	0.0000000000	0.3830480685	0.383917860
18	0.005095375	0.136137967	0.1007316436	0.097334727	0.0738176117	0.0795662399	0.0726417559	0.127515025
19	0.222635083	0.297909893	0.3524384580	0.104164731	0.4299427156	0.3327450070	0.3414460443	0.237250348
20	0.301706663	0.125089910	0.1284247695	0.308245603	0.4347086903	0.3888053358	0.4436670372	0.240371412
21	0.069154108	0.113976665	0.0242586291	0.015337871	0.3421730676	0.1504618376	0.3703937773	0.098565873
22	0.259225923	0.034059406	0.0686228623	0.000000000	0.0586498650	0.0246624662	0.0145454545	0.147794779
23	0.080398439	0.038629493	0.0094958787	0.002816465	0.0548177050	0.0383452624	0.0261362755	0.051781608
24	0.102701210	0.001040750	0.0577526604	0.006125737	0.0276174230	0.0267054054	0.0160210604	0.051037070

> others1

	EWS.T1	EWS.T2	EWS.T3	EWS.T4	EWS.T6	EWS.T7	EWS.T9	EWS.T11	EWS.T12	EWS.T13	EWS.T14	EWS.T15	EWS.T19
[1.]	0.88321121	0.447099938	0.9045713319	0.2642748436	0.744461981	0.560988670	0.501662815	0.74564568	0.644608534	0.52443492	1.000000000	0.586973677	0.6324333465
[2.]	0.009971738	0.011518157	0.0355143777	0.0752946195	0.099877353	0.01753860	0.039460353	0.06999081	0.015890791	0.10803605	0.02996638	0.025169306	0.058923905
[3.]	0.247389999	0.162400050	0.1481423347	0.1198519450	0.1287614900	0.12349656	0.126466604	0.25891916	0.220931365	0.20070663	0.24023388	0.239810150	1.000000000
[4.]	0.100110025	0.017277249	0.0707091751	0.2420036328	0.069315955	0.000000000	0.080416143	0.29764780	0.182140741	0.34100038	0.133885847	0.278936986	0.5584403747
[5.]	0.125397994	0.000000000	0.1951877112	0.2037551107	0.168270467	0.366039041	0.456555480	0.38428426	0.357436166	0.09864039	0.442897307	0.362491464	0.3003822515
[6.]	0.323485711	0.572458026	0.6409043633	0.000000000	0.403397553	0.480313011	0.541776039	0.51357040	0.428066771	0.57389431	0.590015353	0.457134367	1.000000000
[7.]	0.147132170	0.311245695	0.4594645394	1.000000000	0.113921545	0.131471834	0.148992598	0.17531568	0.090027649	0.16779480	0.080710921	0.199144995	0.334916766
[8.]	0.559763314	0.084812623	0.4043885602	0.2601084813	0.281854043	0.148668639	0.490483235	0.37371795	0.305424063	0.45606509	0.166617357	0.369082840	0.3184911243
[9.]	0.272378055	0.162808085	0.5747138290	0.1365370733	0.279906157	0.300118593	0.093508302	0.38259255	0.421883057	0.74809219	0.364055894	0.116169949	0.3340208312
[10.]	0.377856977	0.381033548	0.8455101883	0.4673433021	0.451615402	0.509103587	0.207019447	0.71581312	0.491206322	0.38575967	0.477492833	0.169287983	0.6860618269
[11.]	0.386477121	0.286201533	0.3311413681	0.0084586210	0.123468980	0.161204944	0.218887282	0.40737810	0.190400029	0.31946301	0.137930094	0.080957188	0.1845616524
[12.]	0.047615575	0.380997521	0.2601720869	0.0731369404	0.328277672	0.341111273	0.101866706	0.31974624	0.103179233	0.21386904	0.259807496	0.115429488	0.7411404404
[13.]	0.414034596	0.351180055	0.4425730093	0.0285384131	0.217347153	0.275124308	0.607955739	0.33524757	0.316303663	0.26094264	0.217837384	0.108318601	0.2865746901
[14.]	0.160339738	0.646926304	0.4843596250	0.4597079087	0.251618416	0.770236677	0.104174219	0.36306904	0.519731731	0.53136559	0.061396240	0.334843855	0.4705577710
[15.]	0.6247886-01	0.066067782	1.3044846-01	0.052099812	0.086047617	0.124152838	0.227062324	0.1078183085	0.1498883471	0.2722835-01	0.087577555	0.050481442	
[16.]	0.502451329	0.117308512	1.0000000000	0.6504574392	0.130204633	0.205023447	0.158910047	0.22381697	0.150419213	0.20981953	0.186940858	0.489164417	0.2225735399
[17.]	0.379928817	0.290550268	0.3278513157	0.2390226346	0.333220027	0.565874190	0.509737663	0.58658922	0.542829721	0.34961609	0.749020235	0.605846601	0.7075401904
[18.]	0.368042853	0.347792004	0.5772145284	0.4615887118	0.189443428	0.098379932	0.338385158	0.22942252	0.080219493	0.60765613	0.121374445	0.203292396	1.806898354
EWS.C8 EWS.C3 EWS.C2 EWS.C4 EWS.C6 EWS.C9 EWS.C7 EWS.C1 EWS.C11 EWS.C10 NB.C1 NB.C2													
[1.]	2.821431e-01	0.2397553689	2.413900e-01	0.720957105	0.284735922	0.290400767	0.426667042	0.5914830055	0.0875091596	2.948255e-01	0.333352122	0.330843808	
[2.]	7.876073e-02	0.0760944915	2.127660e-02	0.022982989	0.068390971	0.062230043	0.003732736	0.0202100997	0.0246360582	4.991201e-02	0.093221753	0.041753319	
[3.]	7.278781e-02	0.0572037099	6.456155e-02	0.199945099	0.093968884	0.089178348	0.072690404	0.1736812744	0.0364204691	7.081315e-02	0.223038847	0.094966040	
[4.]	2.624788e-01	0.066067782	1.3044846-01	0.052099812	0.086047617	0.124152838	0.227062324	0.1078183085	0.1498883471	0.2722835-01	0.087577555	0.050481442	
[5.]	3.215968e-01	0.1268725355	4.720052e-02	0.012930920	0.037959079	0.093682651	0.251070925	0.1656541289	0.1997463482	0.2715050e-01	0.126142985	0.093735865	
[6.]	1.363974e-01	0.1598484473	2.0834849e-01	0.613293051	0.334876925	0.292506562	0.308355208	0.5199593878	0.1871625972	1.991977e-01	0.210489822	0.216767140	
[7.]	2.983019e-01	0.1878249096	2.574516e-01	0.335391680	0.388235760	0.374341923	0.143965483	0.1859636623	0.6583936983	4.458695e-01	0.242370265	0.338993785	
[8.]	0.004773e-01	0.3409270217	5.284424e-01	0.167899486	0.411489152	0.427564103	0.357100592	0.4398951887	0.5187869822	4.541420e-01	0.522485207	0.585305720	
[9.]	1.459214e-02	0.2373414458	3.285552e-01	0.409044034	0.330024750	0.317649789	0.113024647	0.5241569585	0.3938311443	2.160978e-01	0.190342374	0.364391049	
[10.]	7.000000e+00	0.2271635547	3.031688e-01	1.0000000000	0.123576354	0.115983575	0.122104284	0.5101107926	0.1140466414	5.834044e-02	0.574339506	0.300302162	
[11.]	3.535976e-01	0.2087369369	1.061421e-01	0.169663565	0.171888455	0.971404404	0.211792955	0.1699091380	1.0000000000	0.2353683e-01	0.043684684	0.118147835	
[12.]	5.505323e-02	0.2231296485	4.239255e-02	0.689295610	0.132638180	0.124908552	0.017135774	0.2406300131	0.5166982645	6.431384e-02	0.621546416	0.209858575	
[13.]	5.245465e-01	0.3795734841	4.503116e-01	0.115694376	0.81401359	1.0000000000	0.26008073	0.2334547237	0.7320190490	2.617480e-01	0.036697248	0.0876812	
[14.]	8.985447e-02	0.0254544513	2.441866e-02	0.528639495	0.022528355	0.021855093	0.013465223	0.0058521933	0.0823191258	6.525454e-03	0.015148376	0.017235258	
[15.]	5.041975e-02	0.0611494710	1.485271e-01	0.432765387	0.051264219	0.046843177	0.170930406	0.1208583776	0.0000000000	3.988873e-02	0.262431076	1.000000000	
[16.]	1.476837e-01	0.3617663777	2.984226e-01	0.222999588	0.140969163	0.140116527	0.107005826	0.2546894984	0.2911041637	2.335868e-01	0.706018190	0.323824073	
[17.]	3.740802e-01	0.3726905549	0.041470e-01	0.255878589	0.276523694	0.307246261	0.51927117	0.5349517076	0.4302267456	0.224686e-01	0.180038791	0.281232504	
[18.]	2.343872e-01	0.1249020120	1.140580e-01	0.102822054	0.156127515	0.149333682	0.056441077	0.0083616410	0.1524692971	1.089626e-01	0.152338664	0.202116540	
NB.C3 NB.C6 NB.C12 NB.C7 NB.C4 NB.C5 NB.C10 NB.C11 NB.C9 NB.C8 RMS.C4 RMS.C3													
[1.]	0.201792458	0.1838960600	0.4613888732	0.359872611	0.1660560284	0.147285948	0.32632663	0.3399188321	2.458711e-01	0.137309058	0.767234091	0.57006369	
[2.]	0.049538740	0.0172239108	0.0892123927	0.041273396	0.015157684	0.041060097	0.04276491	0.004126593	4.185997e-02	0.07720962	0.01678131	0.15904156	
[3.]	0.197394869	0.0865218585	0.0584604759	0.030735582	0.0465505486	0.074735901	0.124916985	0.1081191170	0.959453e-02	0.120560342	0.424011122	0.26154025	
[4.]	0.229978971	0.2114048737	0.2280128385	0.127193182	0.1750053711	0.155272427	0.124921062	0.3086047617	1.682932e-01	0.27437795	0.153293273	0.06496702	
[5.]	0.408654315	0.216890162	0.2401798622	0.182726845	0.1070126737	0.141619292	0.138080583	0.2902184421	2.607913e-01	0.374633047	0.338988763	0.25956737	
[6.]	0.172056857	0.1781982071	0.3571145560	0.543979991	0.2889406174	0.368827696	0.32495171	0.2874052796	2.113268e-01	0.377098707	0.608464167	0.23810113	
[7.]	0.206117005	0.1196611645	0.1415508847	0.224874322	0.2010450065	0.244230693	0.273482959	0.2591932866	0.816491e-01	0.139452965	0.000000000	0.00775692	
[8.]	0.343047337	0.3137080868	0.4557692308	0.363806706	0.7911735700	0.577958580	1.000000000	0.6390532544	4.643491e-01	0.525788955	0.389546351	0.09950690	
[9.]	0.141074559	0.4301072497	0.0223780551	0.537460039	0.2005001547	0.163091678	0.130066000	0.3718418067	2.718367e-01	0.047256884	0.443332990	0.37882850	
[10.]	0.275431936	0.4207019447	0.1465871233	0.347020996	0.7354923685	0.428990200	0.211900519	0.2175563648	3.561633e-01	0.145424963	0.295808476	0.46703339	
[11.]	0.175038882	0.4306256651	0.1800596278	0.326755628	0.3351529616	0.041118612	0.46691878	0.4193293132	2.859832e-01	0.305028787	0.094763841	0.16515151	
[12.]	0.182295464	0.1835350736	0.0236254922	0.488551845	0.8142773808	0.480020417	0.046594721	0.1535656993	8.677264e-02	0.299620825	0.329226509	0.33724661	
[13.]	0.218782828	0.2196582394	0.0540654017	0.143322362	0.2053014917	0.043385391	0.251488199	0.1503606695	6.859724e-02	0.130856503	0.240913229	0.35671265	
[14.]	0.024729401	0.036982754	0.0123258584	0.045186183	0.0277331814	0.026490238	0.005178932	0.0000000000	1.450101e-02	0.063571392	0.019328116	0.06206950	
[15.]	0.223585515	0.1154935175	0.2657592767	0.549451095	0.1919924495	0.163727584	0.144503502	0.2105210869	2.404749e-01	0.217872932	0.116735383	0.18096468	
[16.]	0.83182009	0.2061247691	0.1505257922	0.257780304	0.2880133580	0.107574250	0.593363649	0.3885533608	4.698380e-01	0.214722183	0.136777036	0.09748472	
[17.]	0.334189794	0.5330520675	0.4109413741	0.442113893	0.3854574902	0.							

2. EWS-NB-low pattern: EWS 與 NB 兩組的平均，低於 BL 與 RMS 兩組的平均

```
# 2. EWS-NB-low pattern : EWS 與 NB 兩組的平均，低於 BL 與 RMS 兩組的平均
EWS_NB_low_rows <- which(EWS_data_mean < RMS_data_mean & EWS_data_mean < BL_data_mean &
  NB_data_mean < RMS_data_mean & NB_data_mean < BL_data_mean)

# 過濾:只留下 EWS 與 NB 兩組的平均，低於 BL 與 RMS 兩組的平均的資料
EWS_NB_low_data <- data1_norm[EWS_NB_low_rows, ]

target2 <- EWS_NB_low_data[, which(labels == "EWS" | labels == "NB")]
target2_mean <- apply(target2, 1, mean)
target2_sd <- apply(target2, 1, sd)

others2 <- EWS_NB_low_data[, -which(labels == "EWS" | labels == "NB")]
others2_mean <- apply(others2, 1, mean)
others2_sd <- apply(others2, 1, sd)

snr2 <- SNR(target2_mean, target2_sd, others2_mean, others2_sd)
idx2 <- which.min(snr2) # 索引相對於 EWS_NB_low_data
```

3. EWS-RMS-low pattern : EWS 與 RMS 兩組的平均，低於 BL 與 NB 兩組的平均

```
# 3. EWS-RMS-low pattern : EWS 與 RMS 兩組的平均，低於 BL 與 NB 兩組的平均
EWS_RMS_low_rows <- which(EWS_data_mean < BL_data_mean & EWS_data_mean < NB_data_mean &
  RMS_data_mean < BL_data_mean & RMS_data_mean < NB_data_mean)

EWS_RMS_low_data <- data1_norm[EWS_RMS_low_rows, ]

target3 <- EWS_RMS_low_data[, which(labels == "EWS" | labels == "RMS")]
target3_mean <- apply(target3, 1, mean)
target3_sd <- apply(target3, 1, sd)

others3 <- EWS_RMS_low_data[, -which(labels == "EWS" | labels == "RMS")]
others3_mean <- apply(others3, 1, mean)
others3_sd <- apply(others3, 1, sd)

snr3 <- SNR(target3_mean, target3_sd, others3_mean, others3_sd)
idx3 <- which.min(snr3) # 索引相對於 EWS_RMS_low_data
```

畫出所選 gene 在全部樣本上的 normalized 表現量 scatter plot

用四種不同顏色 + 四種不同符號分別代表 EWS / BL / NB / RMS(每組各自一種顏色與一種符號)

```
# 幫四種類別設定顏色跟點類型
```

```
categories <- c("EWS", "BL", "NB", "RMS")
```

```
color <- c(EWS = "blue", BL = "green", NB = "red",  
          RMS = "black")
```

```
pch <- c(EWS = 16, BL = 17, NB = 15, RMS = 18)
```

```
col_color <- color[labels]
```

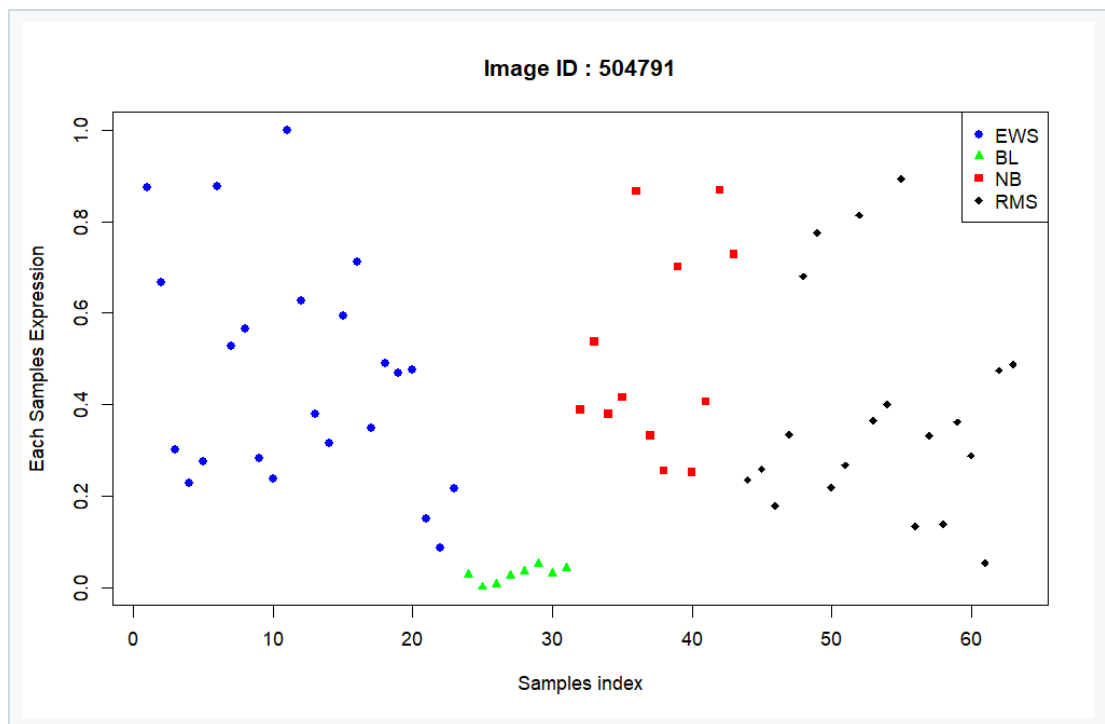
```
col_pch <- pch[labels]
```

```
# picture1
```

```
# 這個 gene 在所有樣本的 normalized 值
gene_expr1 <- data1_norm[bl_low_rows[idx1], ]
plot(gene_expr1, col = col_color, pch = col_pch,
     xlab = "Samples index",
     ylab = "Each Samples Expression",
     main = paste("Image ID :", Image_id[bl_low_rows[idx1]]))

legend("topright", legend = categories,
     col = color[categories], pch = pch[categories])
```

Output:

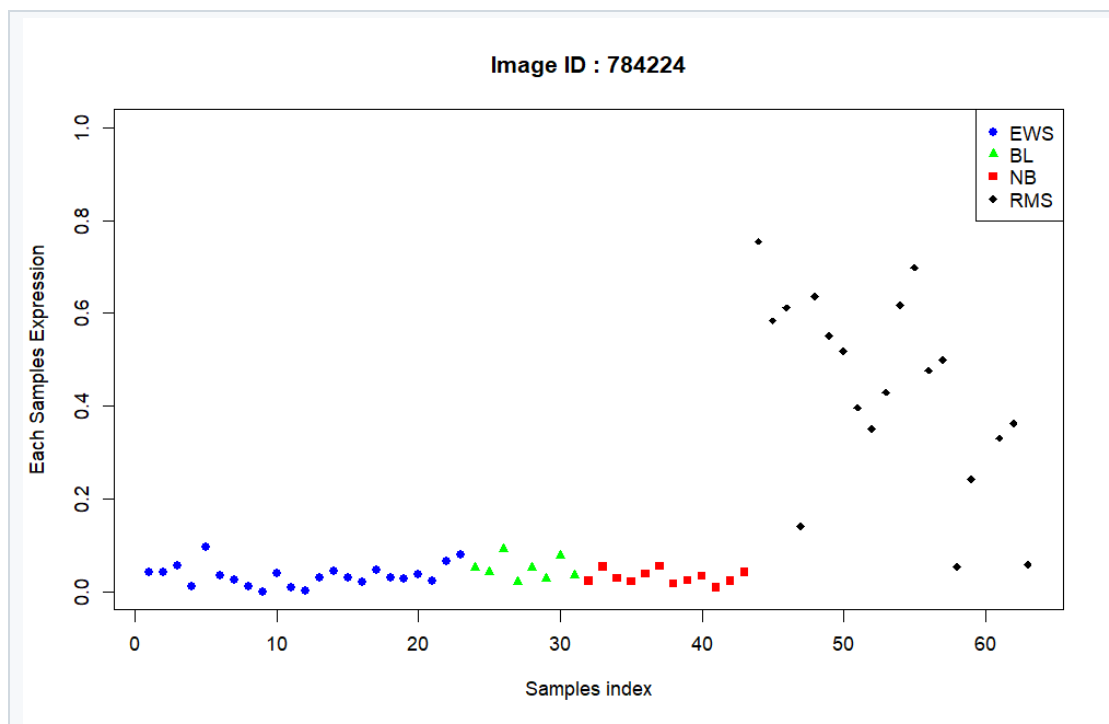


```
# picture2
```

```
# 這個 gene 在所有樣本的 normalized 值
gene_expr2 <- data1_norm[EWS_NB_low_rows[idx2], ]
plot(gene_expr2, col = col_color, pch = col_pch,
     xlab = "Samples index",
     ylab = "Each Samples Expression",
     main = paste("Image ID :",
                  Image_id[EWS_NB_low_rows[idx2]]))

legend("topright", legend = categories,
      col = color[categories], pch = pch[categories])
```

Output:

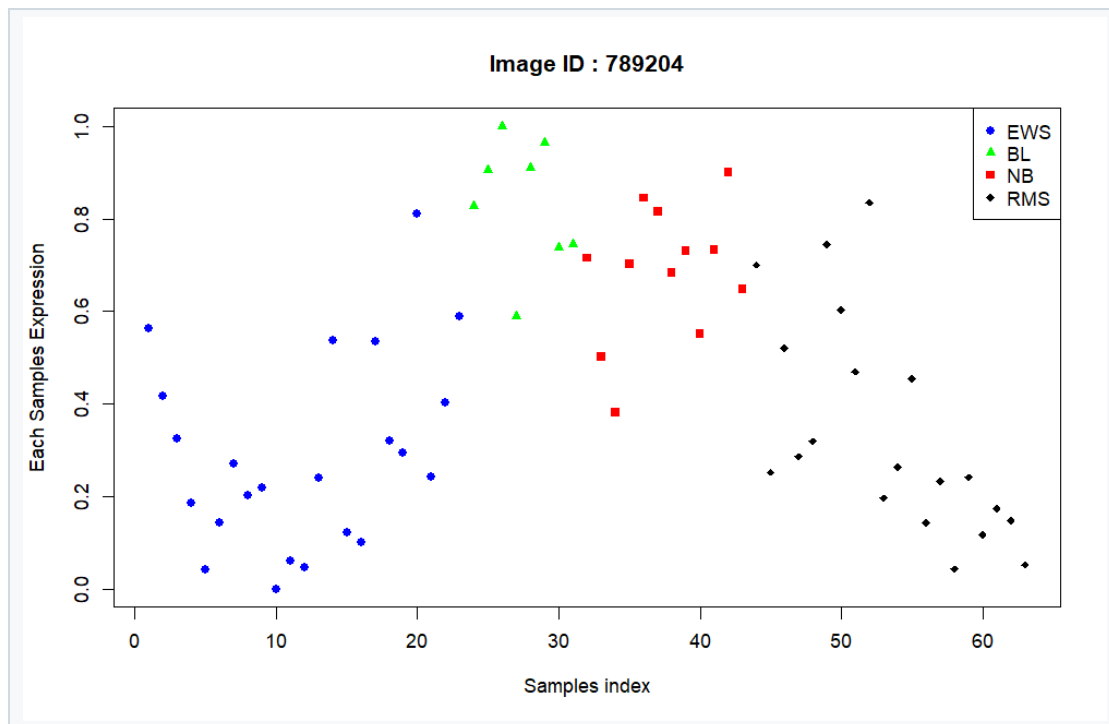



```
# picture3
```

```
# 這個 gene 在所有樣本的 normalized 值
gene_expr3 <- data1_norm[EWS_RMS_low_rows[idx3], ]
plot(gene_expr3, col = col_color, pch = col_pch,
     xlab = "Samples index",
     ylab = "Each Samples Expression",
     main = paste("Image ID :",
                  Image_id[EWS_RMS_low_rows[idx3]]))

legend("topright", legend = categories,
      col = color[categories], pch = pch[categories])
```

Output:



加分題 B

用 `par(mfrow = ...)` 把主題目第 3、4 點的四張 `scatter plot` 排成一張組合圖，

但不要用方方正正的 `2x2` 等分排版。請自己設計版面

想法：我想把第四點的兩張圖放在第一列（大圖），三張個別個案例在第二列（小圖）

```
# +-----+-----+-----+
# | 1      | 2      |      ← 第 4 點的兩張,較寬
# +-----+-----+-----+
# | 3      | 4      | 5      |      ← 第 3 點的三張
# +-----+-----+-----+
# 準備想要視覺化的資料
plot_data1 <- rbind(data1_norm[which(Image_id == 770394), ],
                    data1_norm[which(Image_id == 814260), ],
                    data1_norm[which(Image_id == 491565), ])

# - Perform scatter plots for Image ID “770394” vs “236282”

Image_id_770394 <- data1_norm[which(Image_id == 770394), ]
Image_id_236282 <- data1_norm[which(Image_id == 236282), ]

# - Perform scatter plots for Image ID “812105” vs “784224”
Image_id_812105 <- data1_norm[which(Image_id == 812105), ]
Image_id_784224 <- data1_norm[which(Image_id == 784224), ]

# AI
layout(matrix(c(1, 1, 1, 2, 2, 2,
                3, 3, 4, 4, 5, 5), nrow = 2, byrow = TRUE),
        heights = c(2, 1) # 上列高度是下列的 2 倍
    )
```

```

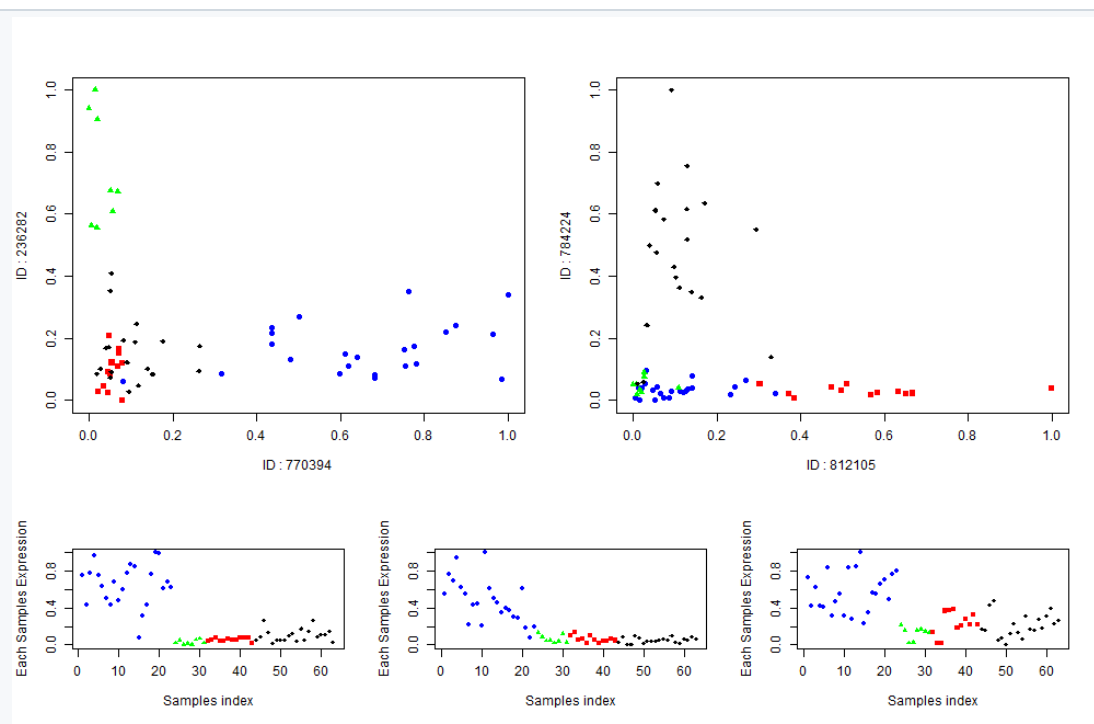
plot(Image_id_770394, Image_id_236282,
     col = col_color, pch = col_pch,
     xlab = "ID : 770394", ylab = "ID : 236282")

plot(Image_id_812105, Image_id_784224,
     col = col_color, pch = col_pch,
     xlab = "ID : 812105", ylab = "ID : 784224")

# 不同 ID 不同點樣式，不同顏色
for (i in 1:3){
  # x 軸是樣本數，y 軸是基因表數
  plot(plot_data1[i, ], col = col_color, pch = col_pch,
       cex = 0.8,
       xlab = "Samples index",
       ylab = "Each Samples Expression")
}
# 畫完重置，恢復成一個視窗一張圖
layout(1)

```

Output:



AI:

par(mfrow = ...) 可以怎麼玩

`par(mfrow)` 本身其實很單純、也很受限——它只能排「規則的格子」。真正能「玩」出不對稱版面的是它的兄弟 `layout()`。我兩個都講,你就知道界線在哪。

`par(mfrow)` / `par(mfcol)` :只能排規則格子

```
r
par(mfrow = c(2, 3)) # 2 列 3 欄,共 6 格,每格一樣大,由左到右、由上到下填
par(mfcol = c(2, 3)) # 一樣 6 格,但改成由上到下、由左到右填(填的順序不同)
```

`mfrow` = 「multiple figures, row-wise(逐列填)」, `mfcol` = 逐欄填。差別只在圖填進去的順序,格子大小永遠均等。

所以 `par(mfrow)` 能做的變化就是:

```
r
par(mfrow = c(1, 5)) # 五張橫排
par(mfrow = c(5, 1)) # 五張直排
par(mfrow = c(2, 3)) # 2×3(會有一格空著,如果只畫 5 張)
```

每格一樣大,這就是它的極限。做不出「有的圖大、有的圖小」。加分題要「不要方方正正等分」, `par(mfrow)` 其實達不到,你需要 `layout()`。

`layout()` :真正能玩的工具

`layout()` 用一個矩陣描述版面,矩陣裡的數字代表「第幾張圖佔這一格」。相同數字連在一起,就會合併成一個大格。

例子 1:左邊一張大圖,右邊兩張小圖

```
r
layout(matrix(c(1, 2,
                1, 3), nrow = 2, byrow = TRUE))
```

畫出來的區塊:

```
+-----+-----+
|       | 2 |
|  1    +-----+
|       | 3 |
+-----+-----+
```

圖 1 佔左邊一整欄(因為兩格都是 1),圖 2、3 各佔右邊上下。

例子 2:上排 3 小張,下排 2 大張(適合你的 5 張)

```
r
layout(matrix(c(1, 1, 2, 2, 3, 3,
                4, 4, 4, 5, 5, 5), nrow = 2, byrow = TRUE))
```

```
+---+---+---+---+
| 1 | 2 | 3 |   ← 第3點的三張
+---+---+---+---+
|  4 |   5 |   ← 第4點的兩張,較寬
+---+---+---+---+
```

上排三張等寬(各佔 2 格),下排兩張各佔 3 格、比上排寬。這就是不對稱版面了。

例子 3:一張主圖 + 周邊小圖

```
r
layout(matrix(c(1, 1, 2,
                1, 1, 3,
                4, 5, 6), nrow = 3, byrow = TRUE))
```

圖 1 佔左上 2×2 的大塊,其餘 2-6 是小圖環繞。

控制每格的相對大小

`layout()` 還能指定各列/欄的寬高比例,更精細:

```
r
layout(matrix(c(1, 2,
                3, 4), nrow = 2, byrow = TRUE),
        widths = c(2, 1), # 左欄寬度是右欄的 2 倍
        heights = c(1, 2)) # 下列高度是上列的 2 倍
```

`widths` 控制欄寬比例、`heights` 控制列高比例。這樣連「等分的格子」都能拉成不等分。

看看版面長怎樣(超好用)

排版矩陣有點抽象, `layout.show()` 可以在真正畫圖前預覽格子編號和位置:

```
r
layout(matrix(c(1, 1, 2, 2, 3, 3,
                4, 4, 4, 5, 5, 5), nrow = 2, byrow = TRUE))
layout.show(5) # 5 = 你有幾個區塊,會畫出標了編號的空格子
```

跑這行會顯示每個編號區塊的位置,確認版面對了再開始畫真正的圖。強烈建議每次設計新版面都先 `layout.show()` 看一眼。

用法流程

```
r
# 1. 設定版面
layout(matrix(c(1,1,2,2,3,3,
                4,4,4,5,5,5), nrow = 2, byrow = TRUE))

# 2. 依序畫圖,每個 plot() 自動填進下一個編號區塊
plot(...) # → 進區塊 1
plot(...) # → 進區塊 2
plot(...) # → 進區塊 3
plot(...) # → 進區塊 4
plot(...) # → 進區塊 5

# 3. 畫完重置
layout(1)
```

加分題 C

改用 `cycleData_peak.txt` 這份資料(酵母菌細胞週期, 800 genes x 77 個時間點, 最後一欄 peak 是該 gene 的高峰期別)。

```
cycleData_peak <- read.delim("cycleData_peak.txt")
data2 <- as.matrix(cycleData_peak[, -c(1, ncol(cycleData_peak))]) # 只保留數值資料
cols2 <- colnames(data2)
cols2 # 觀察
```

AI

```
labels2 <- sub("\\\\.\\.|[0-9]+$", "", cols2)
table(labels2) # 檢查名稱有沒有分錯
```

Output:

```
> cols2 # 觀察
[1] "c1n3.1" "c1n3.2" "c1b2.2" "c1b2.1" "alpha0" "alpha7" "alpha14" "alpha21" "alpha28" "alpha35" "alpha42" "alpha49" "alpha56"
[14] "alpha63" "alpha70" "alpha77" "alpha84" "alpha91" "alpha98" "alpha105" "alpha112" "alpha119" "alpha126" "alpha133" "alpha140" "alpha147"
[27] "cdc15.80" "cdc15.90" "cdc15.100" "cdc15.110" "cdc15.120" "cdc15.130" "cdc15.140" "cdc15.150" "cdc15.160" "cdc15.170" "cdc15.180" "cdc15.190" "cdc15.200"
[40] "cdc15.210" "cdc15.220" "cdc15.230" "cdc15.240" "cdc15.250" "cdc15.260" "cdc15.270" "cdc15.280" "cdc15.290" "cdc15.300" "cdc15.310" "cdc15.320" "cdc15.330"
[53] "cdc28.60" "cdc28.70" "cdc28.80" "cdc28.90" "cdc28.100" "cdc28.110" "cdc28.120" "cdc28.130" "cdc28.140" "cdc28.150" "cdc28.160" "cdc28.170" "cdc28.180"
[66] "elu60" "elu60" "elu90" "elu120" "elu150" "elu180" "elu210" "elu240" "elu270" "elu300" "elu330" "elu360" "elu390" "elu420" "elu450" "elu480" "elu510" "elu540" "elu570" "elu600" "elu630" "elu660" "elu690" "elu720" "elu750" "elu780" "elu810" "elu840" "elu870" "elu900" "elu930" "elu960" "elu990" "elu1020" "elu1050" "elu1080" "elu1110" "elu1140" "elu1170" "elu1200" "elu1230" "elu1260" "elu1290" "elu1320" "elu1350" "elu1380" "elu1410" "elu1440" "elu1470" "elu1500" "elu1530" "elu1560" "elu1590" "elu1620" "elu1650" "elu1680" "elu1710" "elu1740" "elu1770" "elu1800" "elu1830" "elu1860" "elu1890" "elu1920" "elu1950" "elu1980" "elu2010" "elu2040" "elu2070" "elu2100" "elu2130" "elu2160" "elu2190" "elu2220" "elu2250" "elu2280" "elu2310" "elu2340" "elu2370" "elu2400" "elu2430" "elu2460" "elu2490" "elu2520" "elu2550" "elu2580" "elu2610" "elu2640" "elu2670" "elu2700" "elu2730" "elu2760" "elu2790" "elu2820" "elu2850" "elu2880" "elu2910" "elu2940" "elu2970" "elu3000" "elu3030" "elu3060" "elu3090" 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"elu41940" "elu41970" "elu42000" "elu42030" "elu42060" "elu42090" "elu42120" "elu42150" "elu42180" "elu42210" "elu42240" "elu42270" "elu42300" "elu42330" "elu42360" "elu42390" "elu42420" "elu42450" "elu42480" "elu42510" "elu42540" "elu42570" "elu42600" "elu42630" "elu42660" "elu42690" "elu42720" "elu42750" "elu42780" "elu42810" "elu42840" "elu42870" "elu42900" "elu42930" "elu42960" "elu42990" "elu43020" "elu43050" "elu43080" "elu43110" "elu43140" "elu43170" "elu43200" "elu43230" "elu43260" "elu43290" "elu43320" "elu43350" "elu43380" "elu43410" "elu43440" "elu43470" "elu43500" "elu43530" "elu43560" "elu43590" "elu43620" "elu43650" "elu43680" "elu43710" "elu43740" "elu43770" "elu43800" "elu43830" "elu43860" "elu43890" "elu43920" "elu43950" "elu43980" "elu44010" "elu44040" "elu44070" "elu44100" "elu44130" "elu44160" "elu44190" "elu44220" "elu44250" "elu44280" "elu44310" "elu44340" "elu44370" "elu44400" "elu4
```

1. peak 欄共有 5 種期別:G1 / G2/ M / S/G2 / M/G1 / S

```
# 1. peak 欄共有 5 種期別:G1 / G2/ M / S/G2 / M/G1 / S
## 將五種 peak 類別各自取出
G1 <- which(cycleData_peak[, ncol(cycleData_peak)] == 'G1')
G2_M <- which(cycleData_peak[, ncol(cycleData_peak)] == 'G2/M')
M_G1 <- which(cycleData_peak[, ncol(cycleData_peak)] == 'M/G1')
S <- which(cycleData_peak[, ncol(cycleData_peak)] == 'S')
S_G2 <- which(cycleData_peak[, ncol(cycleData_peak)] == 'S/G2')
```

2. 從每一種期別中，自己挑一個沒有缺值(NA) 的 gene(共挑 5 個)

我挑每類別中沒有缺值的第一筆資料

```
G1_no_Na <- G1[which(apply(is.na(data2[G1, ]), 1, sum) == 0)[1]]
G2_M_no_Na <- G2_M[which(apply(is.na(data2[G2_M, ]), 1, sum) == 0)[1]]
M_G1_no_Na <- M_G1[which(apply(is.na(data2[M_G1, ]), 1, sum) == 0)[1]]
S_no_Na <- S[which(apply(is.na(data2[S, ]), 1, sum) == 0)[1]]
S_G2_no_Na <- S_G2[which(apply(is.na(data2[S_G2, ]), 1, sum) == 0)[1]]
```

檢查挑選的資料是否有缺值，與 peak 類別是否正確

```
sum(is.na(data2[G1_no_Na, ]))
cycleData_peak[G1_no_Na, ncol(cycleData_peak)]
sum(is.na(data2[G2_M_no_Na, ]))
cycleData_peak[G2_M_no_Na, ncol(cycleData_peak)]
sum(is.na(data2[M_G1_no_Na, ]))
cycleData_peak[M_G1_no_Na, ncol(cycleData_peak)]
sum(is.na(data2[S_no_Na, ]))
cycleData_peak[S_no_Na, ncol(cycleData_peak)]
sum(is.na(data2[S_G2_no_Na, ]))
cycleData_peak[S_G2_no_Na, ncol(cycleData_peak)]
```

Output:

```
# 檢查挑選的資料是否有缺值，與 peak 類別是否正確
> sum(is.na(data2[G1_no_Na, ]))
[1] 0
> cycleData_peak[G1_no_Na, ncol(cycleData_peak)]
[1] "G1"
> sum(is.na(data2[G2_M_no_Na, ]))
[1] 0
> cycleData_peak[G2_M_no_Na, ncol(cycleData_peak)]
[1] "G2/M"
> sum(is.na(data2[M_G1_no_Na, ]))
[1] 0
> cycleData_peak[M_G1_no_Na, ncol(cycleData_peak)]
[1] "M/G1"
> sum(is.na(data2[S_no_Na, ]))
[1] 0
> cycleData_peak[S_no_Na, ncol(cycleData_peak)]
[1] "S"
> sum(is.na(data2[S_G2_no_Na, ]))
[1] 0
> cycleData_peak[S_G2_no_Na, ncol(cycleData_peak)]
[1] "S/G2"
```


3. 為這 5 個 gene 各畫一張 scatter plot(表現量隨時間點變化)

4. 用 rainbow() 或自訂色系上色,並加上 legend

上色規則:這份資料的 77 個欄位分屬數種實驗條件,每種條件底下是一系列時間點(alpha 18 欄 / cdc15 24 欄 / cdc28 17 欄 / elu 14 欄 / cln3 2 欄 / clb2 2 欄)。

同一個實驗條件的所有時間點用同一種顏色,不同條件用不同顏色。

```
categories2 <- unique(labels2)

# 設定不同時間點的顏色
color2 <- c( "alpha" = "#003E19", "cdc15" = "#EECB27",
             "cdc28" = "#E13239", "elu" = "#1F1762",
             "cln3" = "#A13E97", "clb2" = "#5BBDC8" )

# 設定不同時間點的點樣式
pch2 <- c("alpha" = 15, "cdc15" = 16, "cdc28" = 17,
          "elu" = 18, "cln3" = 19, "clb2" = 20)

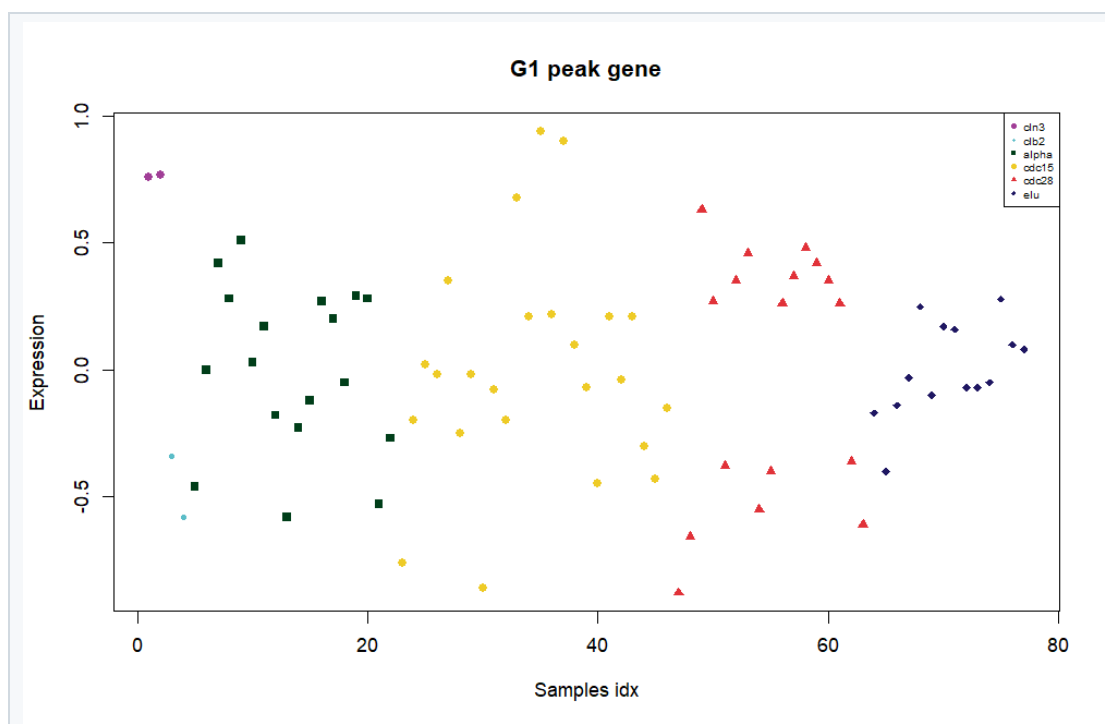
# 套用到所有時間點
col_color2 <- color2[labels2]
col_pch2 <- pch2[labels2]
```

```
# G1
plot(data2[G1_no_Na, ],
     col = col_color2, pch = col_pch2,
     xlab = "Samples idx", ylab = "Expression")

legend("topright", legend = categories2,
     col = color2[categories2], pch = pch2[categories2],
     cex = 0.6)

title("G1 peak gene")
```

Output:

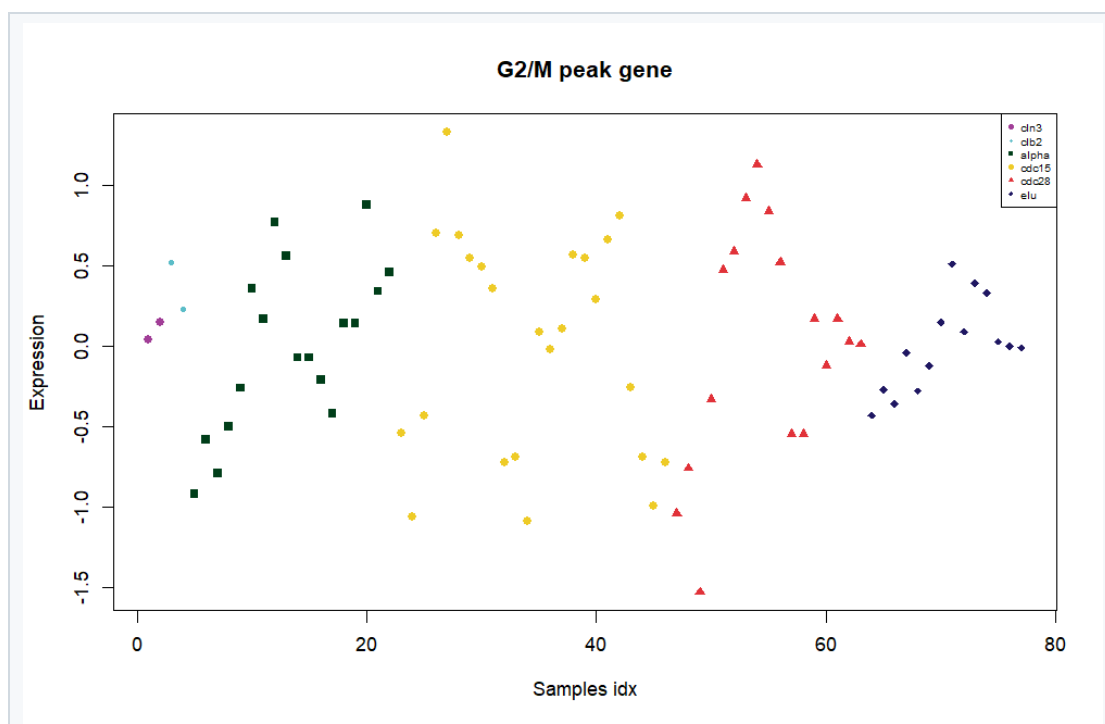


```
# G2/M
plot(data2[G2_M_no_Na, ],
     col = col_color2, pch = col_pch2,
     xlab = "Samples idx", ylab = "Expression")

legend("topright", legend = categories2,
     col = color2[categories2], pch = pch2[categories2],
     cex = 0.6)

title("G2/M peak gene")
```

Output:



```

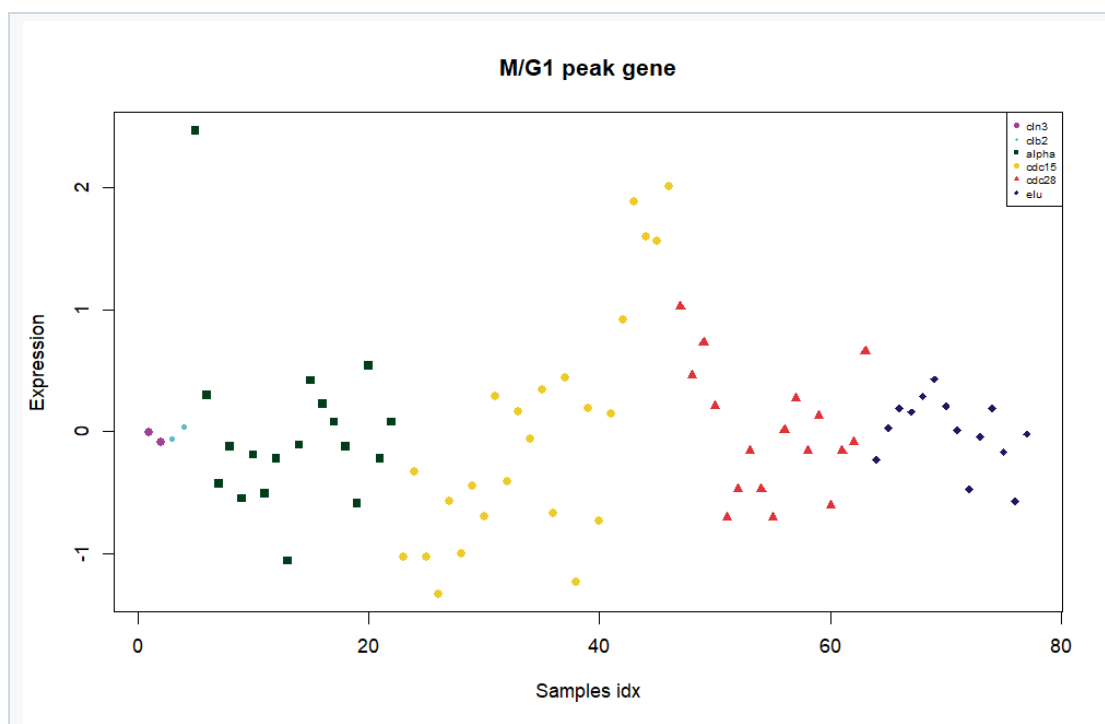
# M/G1
plot(data2[M_G1_no_Na, ],
     col = col_color2, pch = col_pch2,
     xlab = "Samples idx", ylab = "Expression")

legend("topright", legend = categories2,
     col = color2[categories2], pch = pch2[categories2],
     cex = 0.6)

title("M/G1 peak gene")

```

Output:

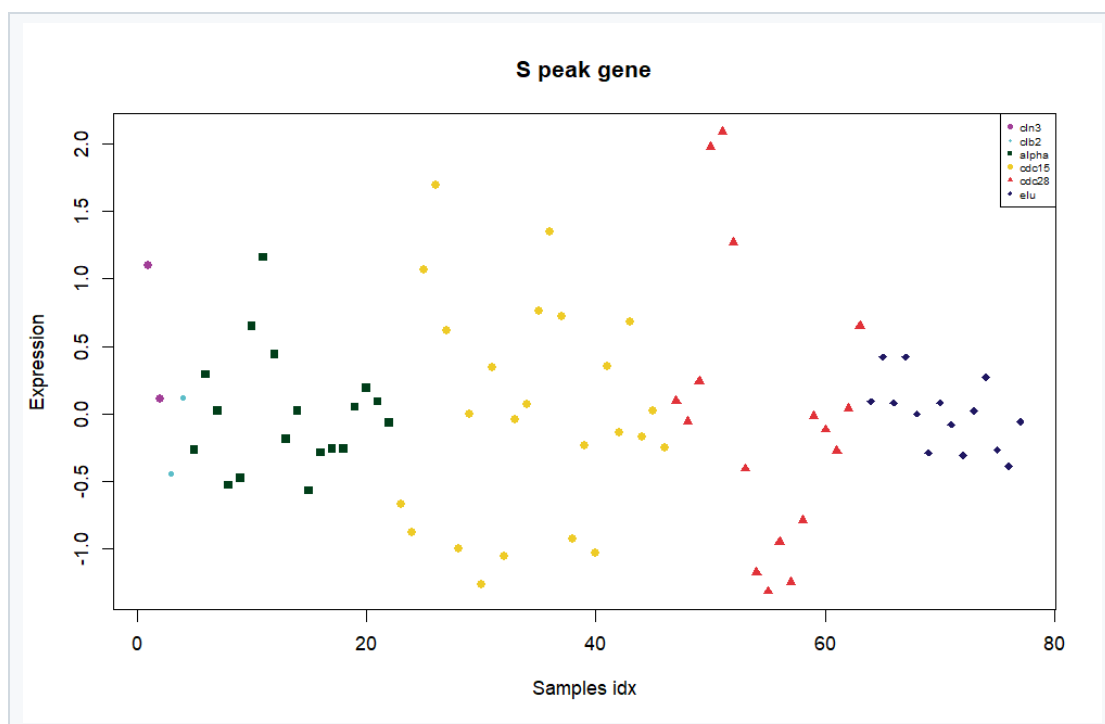


```
# S
plot(data2[S_no_Na, ],
     col = col_color2, pch = col_pch2,
     xlab = "Samples idx", ylab = "Expression")

legend("topright", legend = categories2,
     col = color2[categories2], pch = pch2[categories2],
     cex = 0.6)

title("S peak gene")
```

Output:



```
# S/G2
plot(data2[S_G2_no_Na, ],
     col = col_color2, pch = col_pch2,
     xlab = "Samples idx", ylab = "Expression")

legend("topright", legend = categories2,
     col = color2[categories2], pch = pch2[categories2],
     cex = 0.6)

title("S/G2 peak gene")
```

Output:

